

Downstairs can be bright—but fixture output matters.

Calculation conclusion: build in a useful full-output setting, then dim for evening. The current unselected fixture schedule does not establish that capacity.

RESULT / A CLEAR DESIGN DIRECTION

FAMILY ROOM / 356 SF	DIRECT AVERAGE	WITH FINISH SENSITIVITY
6 broad fixtures / 1000 class	91 lux	98–109 lux
6 broad fixtures / 1500 class	118 lux	127–142 lux
8 broad fixtures / 1500 class	166 lux	178–196 lux

The eight-light option comes close to a 200-lux bright general-light setting without any daylight or sconces. Six stronger broad fixtures can support a softer room, but need a quantified supplemental layer to reach the same brightness.

My recommendation

Use the eight-point broad-beam layout as the brightness benchmark. If a quieter ceiling is more important, retain six and have a real sconce/wall-light package calculated to make up the shortfall. Coordinate all ceiling openings with the acoustic assembly.

What was actually calculated

Manufacturer IES intensity data, an 8-foot ceiling, and point-by-point illumination. The family room uses the full A1-01 356-square-foot area, not the smaller room-label box in the tour.

What “1500 class” means

It is the manufacturer’s nominal package name. The chosen 2700K white-trim IES file integrates to about 1,094 delivered lumens. The calculation uses that actual intensity data, then a 0.80 maintenance factor.

What this is not

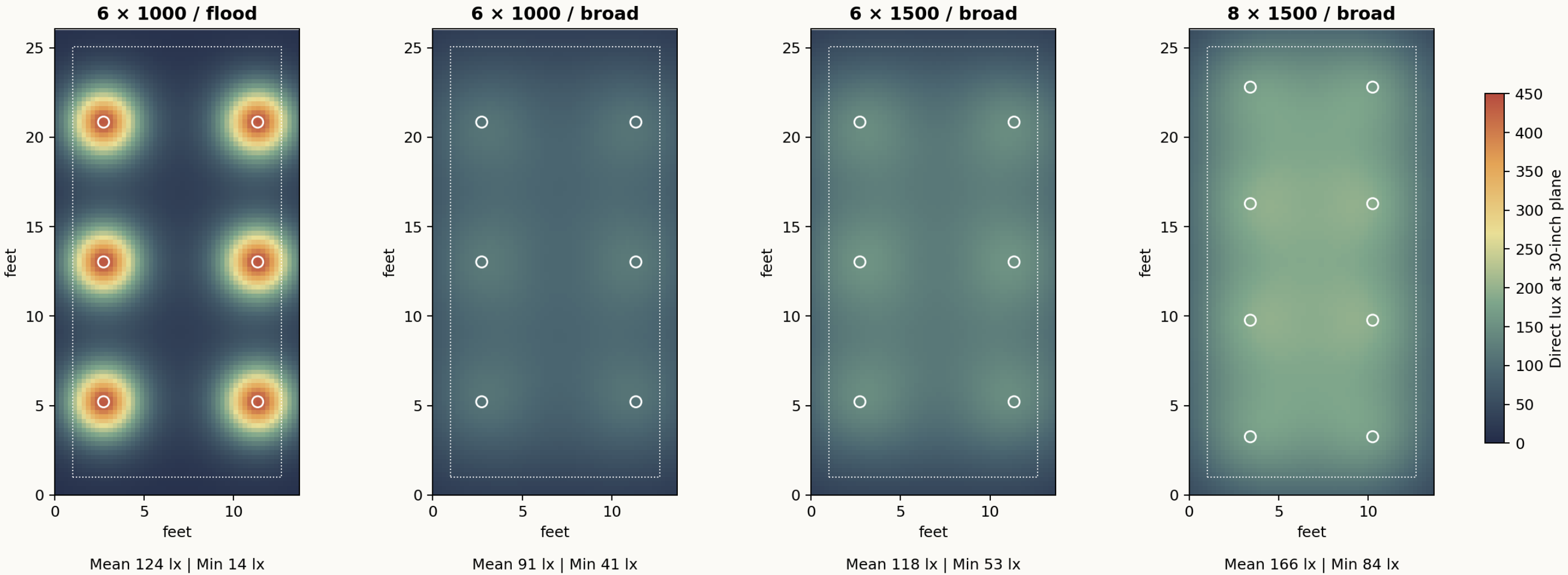
No fixture has been ordered or finally selected. The DMF 4-inch basis is a performance reference; any preferred 2–3-inch substitute needs its own matching photometry.

Sources: A1-01 / A1-03 / A3-04, Aug. 28 set; DMF M Series 2700K GA white-trim IES files. Values are maintained electric-light estimates, not measurements.

Codex | September 25, 2026 | Design study with candidate photometry; final fixtures, placement and construction coordination remain open.

Coverage is as important as the average.

Family-room comparison: the same full room, the same 8-foot ceiling, and a 30-inch evaluation plane. No reflected light is credited in these maps.

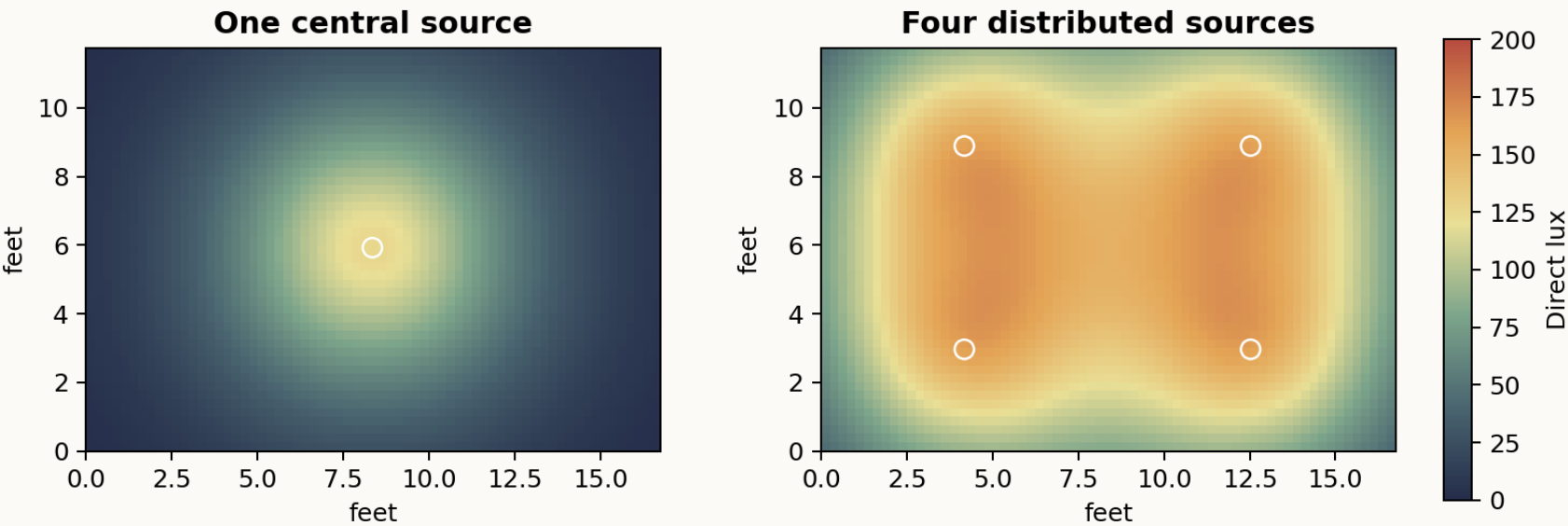


WHAT THE MAPS SHOW	DESIGN CONSEQUENCE
Narrower flood: 14–440 lux in the evaluated zone	Bright pools can coexist with dim gaps. Do not buy on lumen count alone.
Eight broad sources: 84–198 lux before reflections	More even general light; reading, work and the wet-bar counter still need task lighting.

White rings = candidate fixture centres; dotted line = statistical boundary 1 ft inside walls. Common color scale 0–450 lux; 1 foot-candle ≈10.764 lux.

Bedrooms: give the room ambient light and the task its own light.

A central downlight is a poor stand-in for a complete bedroom scheme. Actual fan lights and decorative fixtures remain unselected.



ROOM	TEST BASIS	EST. AVG*
Bedroom 1	4 × 1500 broad	163–181 lx
Bedroom 2	4 × 1000 broad	142–159 lx
Bedroom 3 / WFH	4 × 1500 broad	150–166 lx
Bedroom 4 / office	4 × 1000 broad	155–176 lx
Gym	4 × 1500 broad	198–224 lx

Bedroom 3 example: one central 1500-class downlight averages only 49 direct lux. Four distributed 1500-class sources give 141 direct lux and about 150–166 lux with the tested finishes. A diffuse ceiling fixture could be another solution; its actual IES data must be checked.

*Finish sensitivity: dark to light room assumptions. These are proposed capacity tests, not issued fixture counts. Sources are dimmable; high-output capacity is not the intended bedtime setting.

Aim for 300–500 lux on the activity itself

These are my study targets for reading, desk work and detailed tasks—not a requirement to illuminate the entire bedroom that brightly. Use selected bedside/desk lights and verify the actual working surface.

Areas: A1-01—Bed1 179, Bed2 149, Bed3 198, Bed4 130, Gym 135 sf. Rectangular test proportions are approximate; exact furniture and ceiling placement need coordination.

The remaining first-floor zones need different solutions.

Light the circulation floor, the laundry worktop and faces at mirrors separately. One room-wide brightness number cannot stand in for all three.

ZONE	CALCULATED BASIS	DIRECT / FINISH AVG	CONCLUSION
Hall 101 / breezeway	2 × 1000 broad; floor plane	71 / 82–106 lx	Useful baseline; light finishes help. Confirm final entry/door geometry.
Hall 103	3 × 1000 broad; floor plane	79 / 90–116 lx	Suitable circulation starting point around the 100-lux study target.
Hall 108	2 × 1000 broad; floor plane	73 / 84–112 lx	Similar circulation basis; closet light is a separate task.
Laundry	4 × 1500 broad; 36-inch plane	274 / 296–342 lx	Good general folding/work-surface basis; cabinet shadows may need a task strip.
Bath 1	2 × 1500 broad; 30-inch plane	142 / 154–179 lx	Ambient layer only. Mirror-face and shower/niche output still need actual fixtures.
Bath 2	2 × 1500 broad; 30-inch plane	170 / 184–219 lx	Ambient layer only; use separate mirror lighting and a wet-location fixture layout.
Closet 112	1 × 750 broad; floor plane	33 / 39–57 lx	Floor test is modest; illuminate hanging clothes/shelf fronts with integrated light.
Garage	6 × 1500 downlight surrogates	82 / 87–93 lx	Too little for the 200-lux study target. Select the six shop bars by delivered output.

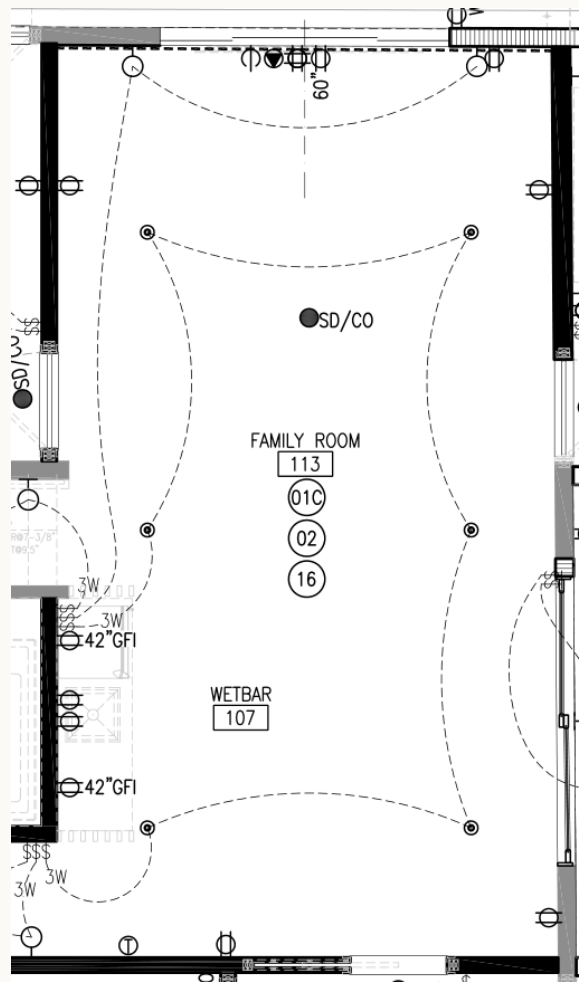
| Garage output budget + uncalculated zones

For 633 sf, six 4,500-lumen shop fixtures estimate about 165–239 average lux using a 0.45–0.65 utilization factor and 0.80 maintenance factor. That is a lumen-method budget, not an IES layout. Stairs, utility closet, mirror faces, shelf fronts and covered patio require their own selected-fixture checks.

Targets are Codex planning criteria, not code findings. Utility-room point layouts are proposals. Garage shop bars, decorative lights and vertical task surfaces are not selected.

A calculation you can hand to your architect and electrician.

The source files, coordinates, assumptions and numerical results are saved beside this report so the fixture supplier can reproduce or replace the model.



PLAN PATTERN / SIX F2 POINTS

The reviewed A1-03 shows six recessed points in the family/wet-bar area. The eight-point option redistributes the layout; it is not simply two extra holes at arbitrary locations. Confirm framing, fan/speaker positions and the resilient-channel detail.

Inputs and assumptions

8-foot ceiling; nominal 2700K; manufacturer Type-C candela data; 0.80 light-loss factor. Horizontal grids: 30 inches for living/bedrooms/baths, 36 inches for laundry, floor level for halls/closet. Quarter-foot direct grid; reported statistics exclude a one-foot perimeter.

Finishes and limits

Supplemental diffuse-box tests use ceiling/wall/floor reflectances of 80/50/20% and 80/20/15%. These are assumptions, not measured paint samples. No daylight, furniture, people, specular reflections or detailed window transmission is modeled. Shadows may lower local light; daylight adds a separate benefit.

Checks completed

On-axis output matches $\text{candela} \div \text{distance}^2$; doubling distance quarters illuminance. Diffuse exchange preserves area reciprocity and limits energy gain. Refining family-room surface patches from 1.5 to 1 foot changed mean light-case output by 0.7%. This checks numerical stability, not field accuracy.

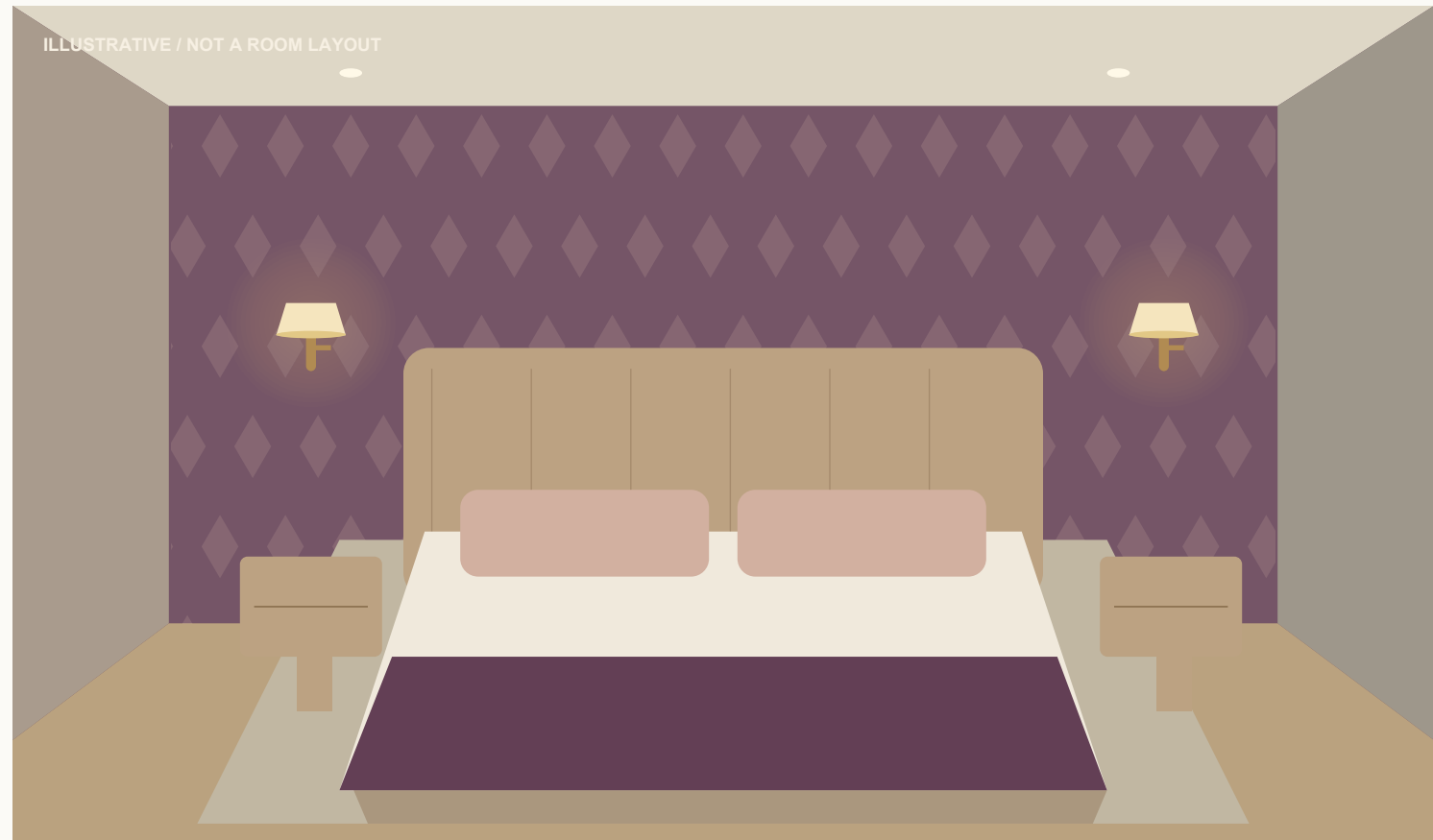
Before final fixture release

Match the chosen housing, trim, color temperature, beam and dimmer to its IES file. Verify the exact room outline and centres; recalculate task planes and shadows; approve a physical mockup. Keep bright-day, evening and night-route layers independently controllable.

SOURCE LINKS: dmflighting.com/product/drd-m-series-4in-downlight/ · A1-01/A1-03/A3-04 in Aug. 28 revision set. Full URLs, IES files, CSVs and assumptions are in README.md and source_manifest.json.

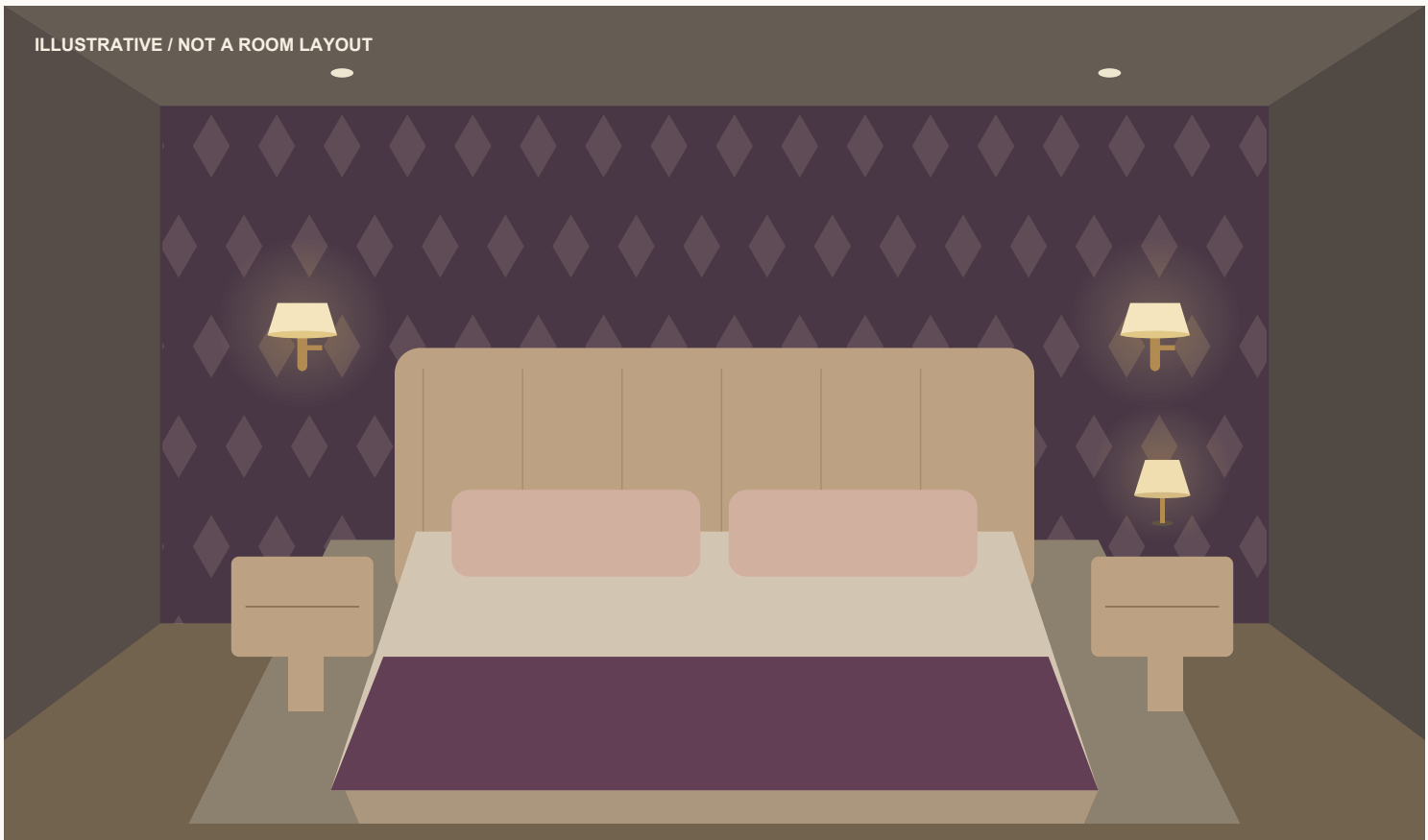
Yes to sconces. Let the bedroom have more than one mood.

Owner update: bedroom wall sconces are planned; portable lamps are possible. Use those layers deliberately, with enough ambient capacity for dressing and cleaning.



01 / BRIGHT + PRACTICAL

Ceiling ambient on; sconces balance the bed wall. This is the useful full-room setting for getting dressed, making the bed and finding things.



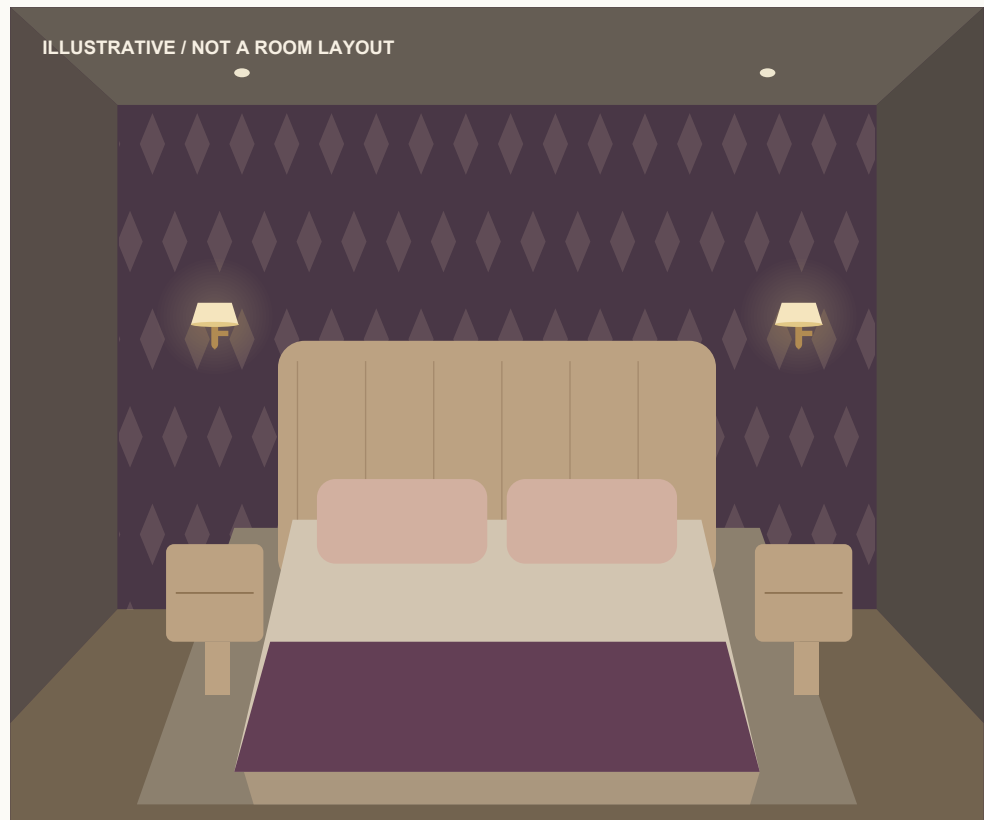
02 / EVENING + PERSONAL

Ceiling ambient low or off; warm shaded sconces carry the room. An optional lamp adds a second pool of light, ideally at a chair or desk rather than duplicating every bedside.

The numerical study still measures the ceiling layer only.
No additional lux has been assigned to these unselected sconces or lamps. They add atmosphere and task opportunities; selected fixture photometry and the actual furniture layout are needed to quantify that contribution.

Three directions for warm, personal bedrooms.

Keep the light warm and the controls consistent. Let the shade, metal and wall finish change the room’s character; these are original concepts, not product selections.



A / PLUM + LINEN

My favorite for the primary

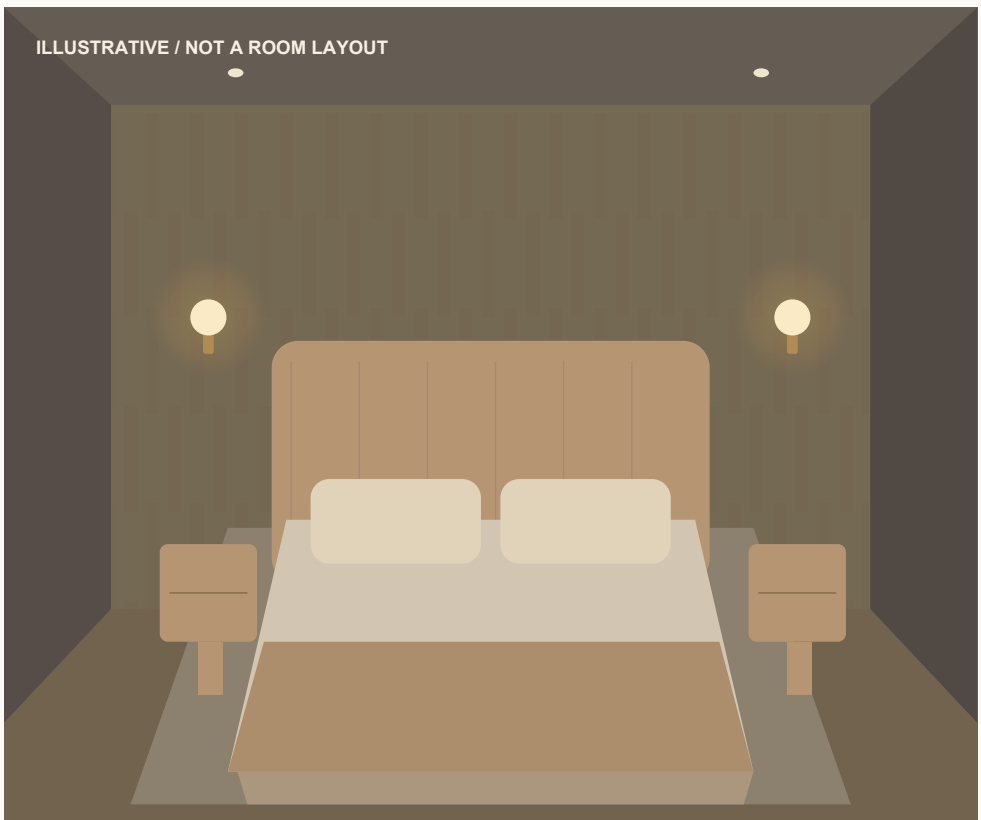
Textured aubergine wall, warm oak headboard, pleated or linen shades and aged brass. Soft pools at eye level; add a reading light only if the chosen shade cannot reach the book.



B / OLIVE + BOTANICAL

A good reading / guest-room direction

Leaf-pattern wallpaper with a quieter bed. Adjustable, shielded sconces let each person aim toward a book. A small lamp at a chair makes another place to settle.



C / SAND + SCULPTURAL

A calmer smaller-bedroom direction

Subtle striped or woven wallpaper, sculptural opal or ceramic sconces and pale oak. Diffuse glow is atmospheric; verify reading performance or provide a task lamp.

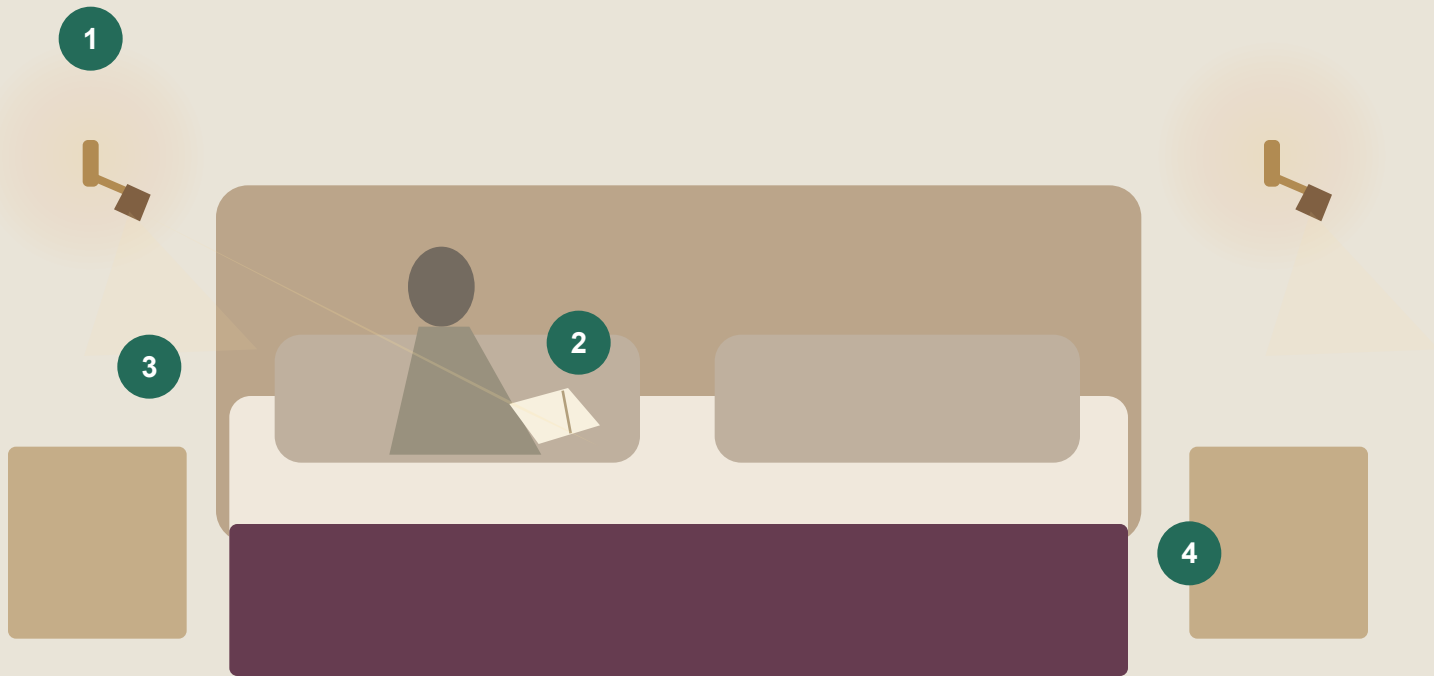
A sconce can be beautiful and still be the wrong reading light.

Choose whether it is decorative glow, directed reading light, or a combination. Test it while sitting in the actual bed: is the page lit without a bright source in either person’s eyes? A dark wallpaper also changes the reflected-light contribution.

Set the bed first. Then place the light and the switch.

The useful detail is the relationship between pillow, eye, book and hand. Fixing every sconce at a generic mounting height would skip that coordination.

BED WALL / COORDINATION DIAGRAM



Final heights and offsets follow a full-size mockup with the chosen headboard.

1 Fixture position

Keep the shade clear of the headboard, artwork and curtains. Check projection and access as well as box center.

2 Reading beam

Aim at the book from a comfortable sitting position. Shield the lamp from both partners' eyes; test left and right separately.

3 Reach + independence

Each person should control their own bedside light without getting up. Keep a useful entry control for the room.

4 Optional lamp power

Provide a coordinated outlet at the intended table, chair or desk. An outlet is not an assumption that a wall dimmer can control any portable lamp.

Give each light layer independent control.

Ceiling ambient, left bedside, right bedside and optional desk / chair light should be usable separately. Local wall controls should work without Wi-Fi. Any added circuit, switch or box gets reconciled against the issued electrical scope before rough-in.

2700K is the proposed bedroom starting point. Confirm the actual lamp / fixture / dimmer combination and low-end behavior in the mockup; the coordination draft's checked 2700K statement specifically covers the Level 2 great room.

Apply the same logic to each bedroom—then personalize it.

The confirmed addition is wall sconces. The examples below translate that direction into usable layers without treating every lamp or fixture count as approved scope.

ROOM	WALL-LIGHT DIRECTION	OPTIONAL LAMP	COORDINATE / VERIFY
Primary 211	Pair of shaded sconces; consider separate reading heads if needed.	Chair / reading corner, if the furniture layout supports one.	Primary is outside the downstairs calculation; verify its own ceiling, headboard and controls.
Bedroom 1	Bedside pair or a bed-specific arrangement after bed size is set.	Table or chair lamp; avoid filling both nightstands by default.	Ceiling test: 163–181 lux.* Balance shade glow with useful reading light.
Bedroom 2	Compact sconces to preserve nightstand space.	Only if a chair / desk creates another use.	Ceiling test: 142–159 lux.* Keep door, closet and fixture projection clear.
Bedroom 3 / WFH	Bedside sconces when used as a bedroom; treat the desk separately.	Adjustable desk lamp for the actual work surface.	Ceiling test: 150–166 lux.* Check screen reflections from seated eye position.
Bedroom 4 / office	Choose the bed / desk arrangement before locking wall boxes.	Task lamp on desk or a shaded guest-room lamp.	Ceiling test: 155–176 lux.* Do not strand a bedside box behind a future cabinet.

*Candidate ceiling-only finish-sensitivity averages, not total room illumination with sconces. No lamp output is credited.

MY RECOMMENDATION

Make sconces part of the architecture, and use portable lamps where they create a useful second place to read, work or unwind. Keep sufficient ceiling capacity for practical tasks; reduce it with dimming when the wall lights take over.

Next drawing to request

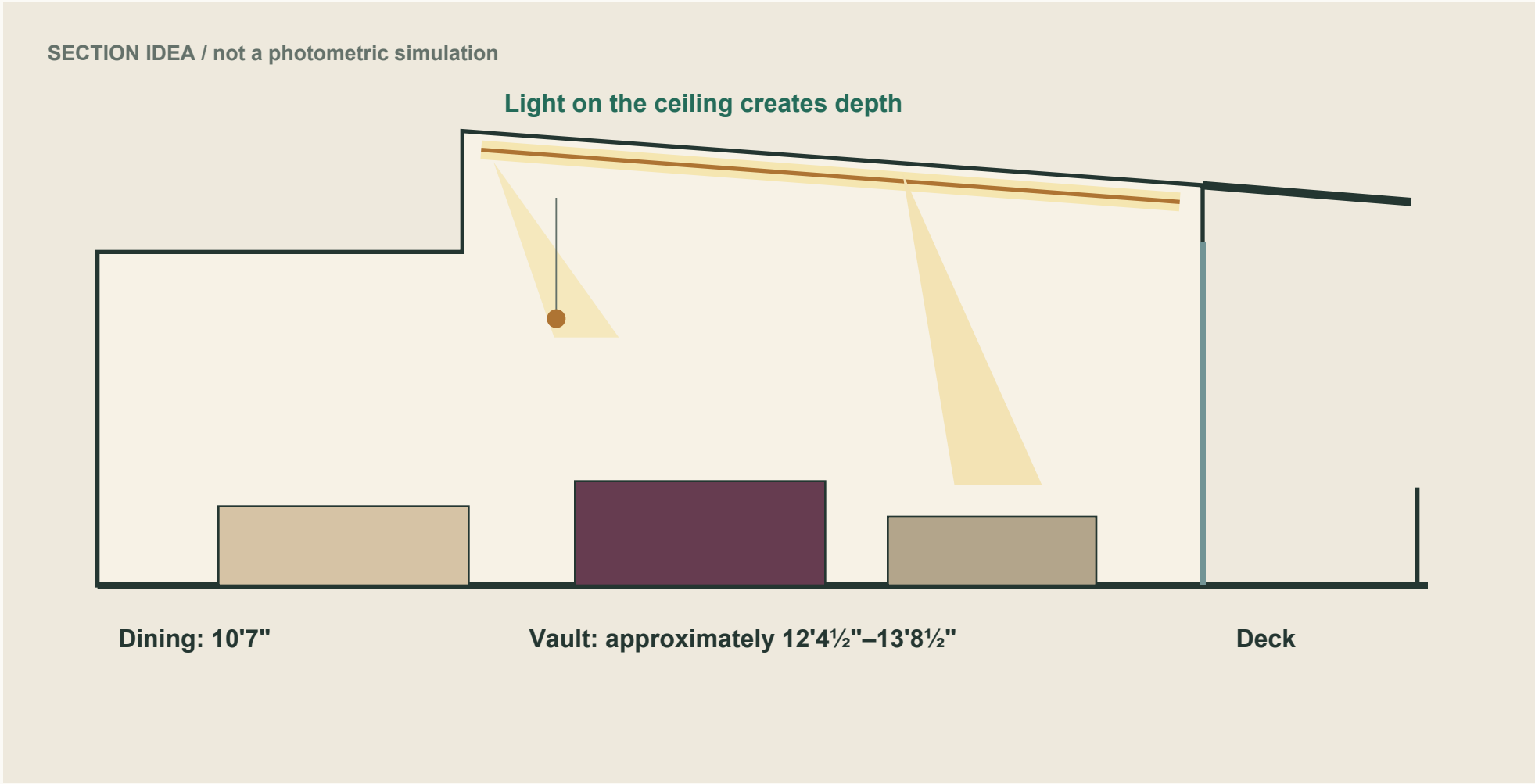
One bed-wall elevation per room, with the actual headboard, fixture, switches and outlets dimensioned together.

Sources: owner direction, September 25; first-floor lighting results.json / reflection_results.json. Styles, scene choices and fixture arrangements are Codex proposals.

Codex | September 25, 2026 | Illustrated design concepts; geometry, light effects and finishes are not measured simulations or final selections.

Let the house glow; keep the ceiling quiet.

Recommendation: invest in concealed light, useful task light and good dimming before adding more decorative fixtures.



Atmosphere comes from layers

Bright counters for cooking; soft walls and ceiling for company; a dim route for late-night movement. One all-on ceiling grid cannot do all three well.

The strongest existing direction

The September 8 coordination package records no island pendants, 2700K visible great-room light and high color rendering. Use that as the study basis; the package itself remains a coordination draft.

The model needs reconciliation

It still shows three island pendants. Its lower-floor ceiling also sits higher than the 8-foot section basis. These images cannot establish final fixture heights or brightness.

Where to spend attention

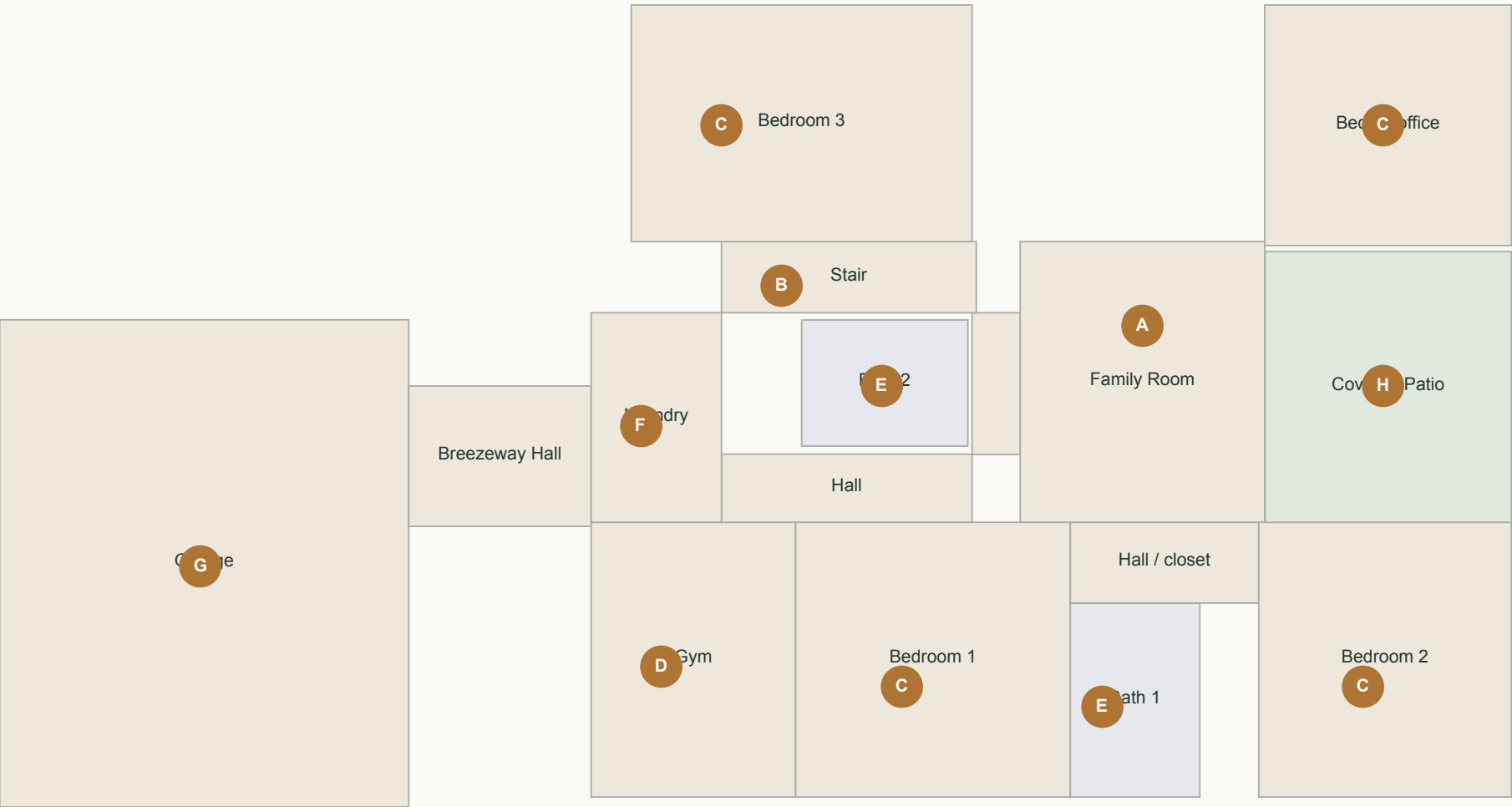
Kitchen task coverage, faces at bathroom mirrors, stair glare, low-level evening light and the deck viewed through glass.

My design judgment

Keep dramatic wallpaper and aubergine stone in the composition. Test them under the actual warm LEDs with the lighter oak before ordering.

Downstairs: comfortable at an eight-foot ceiling.

Lower-floor room coverage. Diagram uses current model footprints; the architectural section, not the model ceiling, controls the 8-foot height assumption.



A / Family room + wet bar

Wash a wall and light the future shelves. Add reading lamps and dimmable task light at the wet bar. Keep light off the TV screen; coordinate any ceiling penetration with the resilient-channel assembly.

B / Stairs, halls + arrival

A continuous, shielded night route matters more than bright ceiling spots. Retain useful general light for cleaning; coordinate the four-step-light baseline with the handrail LED.

C / All four bedrooms

Independent reading light on each side of the bed, a usable desk light and a quiet fan point. Place ambient light away from pillows and monitor reflections. Bed 4 is labeled Office in the tour.

ZONE	LIGHTING APPROACH	DETAIL THAT MATTERS
D / Gym	Diffuse ambient + directed desk/task light	Low ceiling: verify fan and equipment clearances; avoid glare when lying on a bench.
E / Baths 1–2	Face light + shower light + low night layer	Light both sides of the mirror; niche glow does not replace shower illumination.
F / Laundry + closets	Even task light; separate sensor/control	Illuminate washer labels, folding surface and closet fronts, not only the aisle.
G–H / Garage + covered patio	Garage work light; shielded outdoor light	Garage baseline is six shop lights. Patio fan/light listing and corrosion finish need selection.

Sources: current tour room geometry; Aug. 28 A1-03 / A3-04; September 8 lighting and low-voltage packages. Letters are study zones, not fixtures.

Codex design analysis | September 25, 2026 | Dimensions are drawing/model based, not field verified. Not for construction.

Upstairs: give each room its own evening mood.

Warm light connects the open rooms; separate controls let the kitchen work without making the living room feel like a showroom.



A / Kitchen + pantry

No island pendants per the coordination record. Aim task light at the island, sink and cooking surface. Closed uppers now need continuous undercabinet light. Pantry light should reveal shelf contents.

B–C / Living + dining

Living: cove, a lit vertical surface and portable reading lights; no central pendant is my preference, still open in the record. Dining: one statement fixture, with a future-study task-light fallback.

D–F / Arrival + primary suite

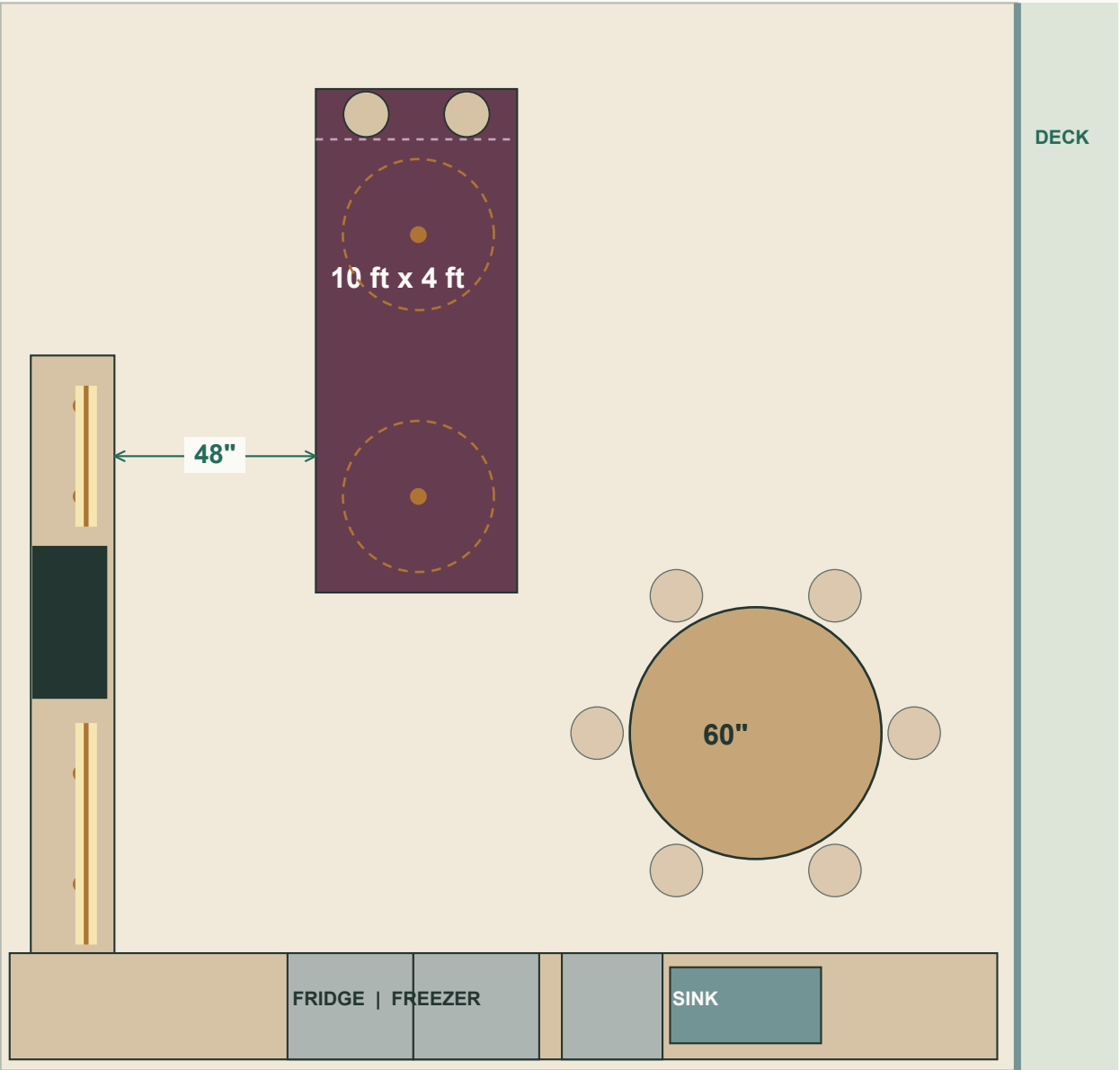
Cove and soft wall light for foyer/hall. Primary: separate bedside reading, cove and fan controls. Primary bath/powder: light faces, then add ambient, shower and night layers.

G / Deck and interior are one nighttime view

Keep exterior sources low and shielded. Bright glass reflections can erase the view; dim the indoor fixtures closest to the sliders.

The kitchen lighting must follow the new kitchen.

Three island pendants are a stale model feature. The September package calls for adjustable downlights, cove and task lighting instead.



Proposed aiming zones; fixture count and beam angles remain unselected.

1 / Island: clear view, useful light

Center the task coverage on the 10-foot footprint, not the former 14-foot run. The two dashed zones are an aiming concept, not an instruction to buy exactly two fixtures.

2 / Uppers change the task-light strategy

Run diffused linear light near the front underside of the closed uppers. Check hood transitions and backsplash reflections; eliminate exposed dots on the aubergine surface.

3 / Refrigerator + sink wall

The moved fridge/freezer creates two separate counter areas. Keep each useful landing lit. Do not wash bright light onto appliance faces while leaving the sink in shadow.

4 / The daily-table fixture stays optional

Do not fix a decorative ceiling box until table size and position pass the occupied-chair test. The island study finds a real circulation pinch here.

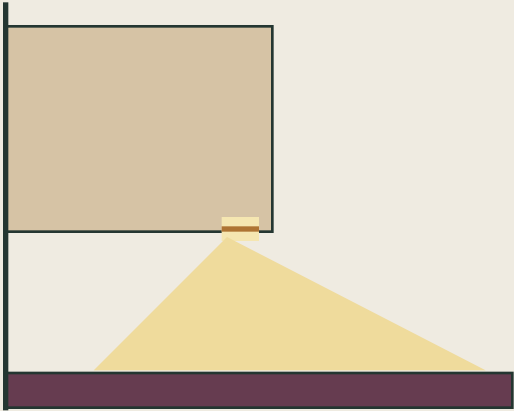
5 / Make one physical mockup

Use actual oak, countertop, backsplash, tape, diffuser, driver and dimmer. Compare full output and the intended dinner scene; a render cannot prove color or glare.

Four details that will determine how the light feels.

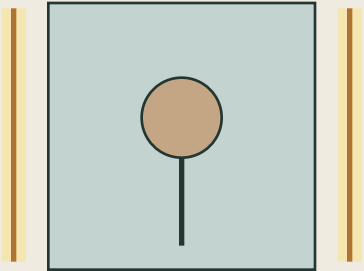
Concept sections. Heights, offsets and source shielding need fixture-specific coordination; these are design principles, not fabrication dimensions.

A / Closed upper cabinet



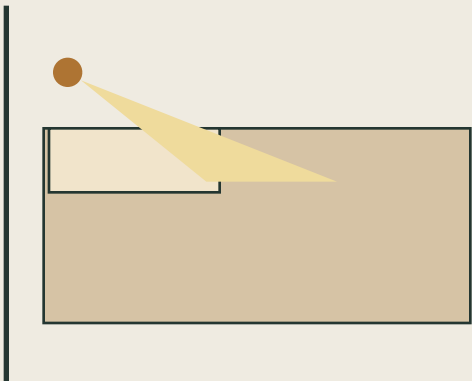
Front-positioned, concealed LED washes the working depth. A diffuser and a small shielding lip hide individual emitters.

B / Bathroom mirror



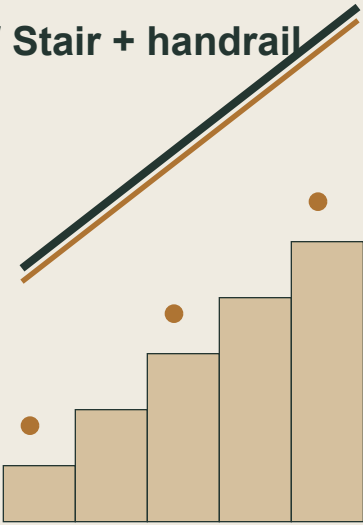
Light faces from the sides. A bright ceiling spot alone can leave eye sockets in shadow; decorative choice must respect mirror and cabinet doors.

C / Bedside reading



Give each sleeper a shielded reading source and reachable control. Keep the shared ambient layer separate; avoid a downlight directly over a pillow.

D / Stair + handrail



Shield the rail light downward. Check seated and standing sightlines. Four step lights plus a rail strip can be redundant: test a low scene before adding more.

Simple controls make the layers usable.

Suggested scenes are relative starting points for commissioning, not measured dimmer percentages or guaranteed light levels.

SCENE	TASK / RECESSED	COVE / WALLS	DECORATIVE / PORTABLE	DECK / ROUTE
Cook + clean	Kitchen bright; other rooms independent	Low–medium	Table low or off	Route visible; grill task light only when cooking
Dinner + company	Kitchen low after prep	Soft, warm background	Table / dining focus; living lamps on	Low rail / wall light; preserve view through glass
Read / work	Local desk or reading source bright	Enough ambient contrast relief	Only useful local sources	Off unless occupied
Movie	No beam on screen; ceiling low/off	Gentle wall/background light	Reading lights only as needed	Very low or off
Night route	General light off	Low shielded path only	Bedrooms independently off	Low path if needed; no floodlighting
All-on / service	Full useful task coverage	Full as needed	Independent override	Maintenance scene with local wall control

Controls: preserve ordinary wall operation

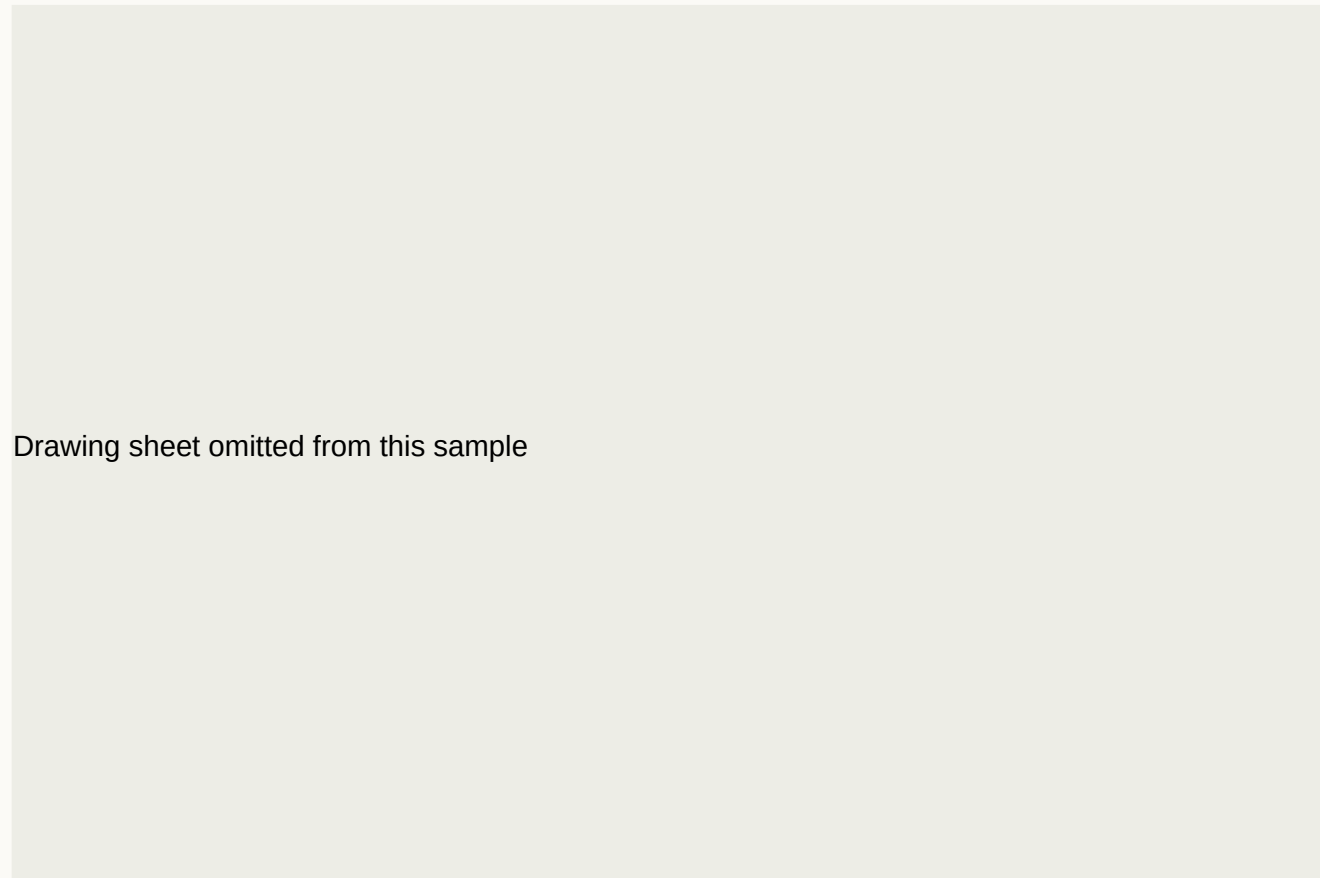
September package selects Caseta-compatible controls and local operation without a network. Verify the exact fixture/driver/dimmer combination; the brand name alone does not guarantee quiet deep dimming.

Resolve one real conflict before purchasing

The older exhibit asks for addressable-capable tape; the newer package emphasizes conventional dimmed layers. Define which runs, if any, actually need addressable control and how they operate from wall controls.

What is established—and what still needs a drawing.

Reconcile the sources before rough electrical. The model is a design tool; its glowing dots are not a fixture takeoff.



Drawing sheet omitted from this sample

A1-04 / AUGUST 28 PROGRESS REVIEW

The source sheet still predates the August 31 kitchen brief. Its kitchen geometry and fixture centers need a coordinated revision. LED intent already exists in the documents; the issue is making the current layers and details agree.

RECONCILE	REQUIRED RESULT
Island pendants	Remove three stale model pendants from the next lighting/model revision; September draft records zero.
Closed uppers / glass cabinets	Replace shelf-era lighting assumptions with actual upper-cabinet shop drawings. Glass-display LED is conditional on a retained display cabinet.
Ceiling planes	Set downstairs to the A3-04 8-foot basis; verify final build-up. Aim vaulted fixtures using issued slope and height.
Fixture + LED count	Arch. / electrician count the revised RCP; separate recessed fixtures from cove runs and decorative points.
Trade coordination	Dimension driver access, cove section, fan support, niche waterproofing and acoustic penetrations.
Exterior / procurement	Select salt-air-suitable finishes and outdoor listings. Confirm grill light and circuit layout; complete SKUs before ordering.

SOURCE REGISTER: Aug. 28 revision set A1-03, A1-04, A3-04; Sept. 8 Lighting, Controls & LED Signoff (coordination draft, not signed release); Sept. 8 Low-Voltage Prewire v3; Aug. 12 Lighting Exhibit; Aug. 31 sent kitchen brief; Sept. 25 tour_geometry.json. Full paths and links are in sources.md beside these PDFs.